**139. There are three cans A, B and C. The capacities of A, B and C are 6 litres, 10 litres and 16 litres respectively. The can C contains 16 litres of milk. The milk has to be divided in them using these three cans only.**

**Consider the following statements:**

1. **It is possible to have 6 litres of milk each in can A and can B.**
2. **It is possible to have 8 litres of milk each in can B and can C**

**Which of statements given above is/are correct?**

(a) 1 only (b) 2 only
(c) Both 1 and 2 (d) Neither 1 nor 2

**Ans. (a) :** The first statement is correct, since the filling capacity of vessel A and vessel B is 6 liters and 10 liters respectively, so it is possible to take 6 liters of milk in vessel A and vessel B each. But, the second statement is not correct, because the filling capacity of vessel B and vessel C is 10 liters and 16 liters respectively and vessel C contains 16 liters of milk. Therefore, although it is possible to take 8 liters of milk in vessel B but not in vessel C, because 16 liters of milk is already present in vessel C.

**140. There are four persons A, B, C, D and A has some coins. A gave half of the coins to B and 4 more besides. B gave half of the coins to C and 4 more besides. C gave half of the Coins to D and 4 more besides. Both B and D end up with same number of coins. How many coins did A have originally?**

(a) 96 (b) 84
(c) 72 (d) 64

**Ans. (c) :** Let there are $x$ coin originally with A
According to question
A gave half + 4 coins to B

$\therefore$ coins with $B = \frac{x}{2} + 4$

again B gave half of its coins + 4 to C

$\therefore$ coins with $C = \frac{1}{2}\left(\frac{x}{2}+4\right)+4$

and coins left with $B = \left(\frac{x}{2}+4\right) - \left[1/2\left(\frac{x}{2}+4\right)-4\right]$

$= \frac{1}{2}\left(\frac{x}{2}+4\right) - 4 = \frac{x}{4} - 2$

Now C gave half of its coins + 4 coins to D

$\therefore$ coins with $D = \frac{1}{2}\left[\frac{1}{2}\left(\frac{x}{2}+4\right)+4\right]+4 = \frac{x}{8}+7$

According to question now B and D have same no. of coins

$\therefore \frac{x}{4} - 2 = \frac{x}{8} + 7 \Rightarrow \frac{x}{8} = 9$

$\therefore \quad x = 9 \times 8 = 72$

**141. How many three-digit numbers can be generated from 1, 2, 3, 4, 5, 6, 7, 8, 9 such that the digits are in ascending order?**

(a) 80 (b) 81
(c) 83 (d) 84

**Ans. (d) :** Any 3 numbers out of 9 can be selected in $^9C_3$ ways.
Now, these three numbers can be arranged among themselves in ascending order in only 1 way.
Hence, total no. of ways = $^9C_3 \times 1 = 84$.

**142. In a carrom board game competition, m boys and n girls (m>n>1) of a school participate in which every student has to play exactly one game with every other student. Out of the total games played, it was found that in 221 games one player was a boy and the other player was a girl.**

**Consider the following statements:**

1. **The total number of students that participated in the competition is 30.**
2. **the number of games in which both players were girls is 78.**

**Which of the statements given above is/are correct?**

(a) 1 only (b) 2 only
(c) Both 1 and 2 (d) Neither 1 nor 2

**Ans. (c) :** Since in 221 games each boy plays exactly one game with each girl so

$m \times n = 221 \quad \ldots\ldots\ldots(i)$

also $m > n > 1$ $\therefore$ m must be 17 and n = 13 to satisfy the equation 1.

$\therefore$ total no. of students = m + n = 17 + 13 = 30

Again no. of games in which both players are girl = $^{13}C_2 = 78$

Hence both the statements are correct.

**143. A person has 4 coins each of different denomination. What is the number of different sums of money the person can form (using one or more coins at a time)?**

(a) 16 (b) 15
(c) 12 (d) 11

**Ans. (b) :** No. of different sums of money = any 1 coin at a time + any 2 coins + any 3 coins in all 4 coins
$= {}^4C_1 + {}^4C_2 + {}^4C_3 + {}^4C_4 = 4 + 6 + 4 + 1 = 15.$

**144. How many numbers lie between 300 and 500 in where 4 comes only one time?**

(a) 99 (b) 100
(c) 110 (d) 120

**Ans. (a) :** Numbers between 300 and 500 in where 4 occurs only once
300-399 = 18
400-410 = 10
411-420 = 9
421-430 = 9



431-439 = 8
450-460 = 10
461-470 =9
471-480 = 9
481-490 = 9
491-500 = 8
Required numbers are - 18+ 10 + 9 + 9 + 8 + 10 + 9 + 9 + 9 + 8 = 99.

**145. In an examination, there are three subjects A, B and C. A student has to pass in each subject. 20% students failed in A, 22% students failed in B and 16% failed in C. The total number of students passing the whole examination lies between.**

(a) 42% and 84% (b) 42% and 78%
(c) 58% and 78% (d) 58% and 84%

**Ans. (a) :** Pass % is minimum when failure % is maximum.
Maximum % of failure = 20 + 22 + 16 = 58%
$\therefore$ Minimum pass % = 100 – 58 = 42%
Now, pass % is maximum, when failure % is minimum. Minimum % of failure = 16%
($\because$ 22 % failure covers both 16% and 20% i.e., 22% is the minimum value of failure %).
$\therefore$ Maximum % of passing = 100 – 16 = 84%
Hence, pass % range = 42% to 84%.

**146. How many letters of the English alphabet (capitals) appear same when looked at in a mirror?**

(a) 9 (b) 10
(c) 11 (d) 12

**Ans. (c) :** The letters (capital letters) of the English alphabet are-
ABCDEFGHIJKLMNOPQRSTUVWXYZ
The following letters look the same when viewed in the mirror,
AHIMOTUVWXY
Required number of letters = 11

**147. There are 240 balls and n number of boxes $B_1$, $B_2$, $B_v$, ..... $B_n$. The balls are to be placed in the boxes such that $B_1$ should contain 4 balls more than $B_2$' $B_2$ should contain 4 balls more than $B_2$. $B_2$ should contain 4 balls more than $B_3$' and so on. which one of the following cannot be the possible value of n?**

(a) 4 (b) 5
(c) 6 (d) 7

**Ans. (d) :** Common difference = 4
$\therefore$ it can be considered as Arithmetic Progression.
$B_2 - B_1 = B_3 - B_2 \ldots\ldots = 4$
Total no. of balls = 240

$\therefore \frac{n}{a}(2a + (n-1)d) = 240$

$\Rightarrow 2a - (n-1) + 4 = \frac{480}{n}$

Since left hand side of the equation gives integer value, right hand side of the equation should give an integer value.
Thus 480 should be divisible by n.
Out of the given option 480 is not divisible by 7
$\therefore$ 7 cannot be the possible value of n.

**148. Four metal rods of lengths 78 cm, 117 cm and 169 cm are to be cut into parts of equal length. .Each part must be as long as possible. What is the maximum number of pieces that can be cut?**

(a) 27 (b) 36
(c) 43 (d) 480

**Ans. (b) :**The given lengths of four metal rods are 78 , 104 , 117 and 169 cm
Now 78 = 13x2x3
104 = 13 x 2 x 2 x2
117 = 13 x 3 x3
169 = 13 x 13
The maximum length of each part is as possible.
= LCM = 13 cm
: . Number of pieces = 6 + 8 + 9 + 13 = 36

**149. While adding the first few continuous natural numbers, a candidate missed one of the numbers and wrote the answer as 177. What was the number missed?**

(a) 11 (b) 12
(c) 13 (d) 14

**Ans. (c) :** Let there be n natural numbers and $x$ be the missed number.

$\therefore \frac{n(n+1)}{2} - x = 177$

$\Rightarrow n(n+1) - 2x = 354$

$\Rightarrow n(n+1) = 354 + 2x \quad \ldots\ldots\ldots(1)$

Out of the given options only 13 as value of $x$ satisfies the equation (1) as
19 (19 + 1) = 354 + 2 (13) = 380
Hence option (c) is correct LHS = RHS

**150. How many time are an hour hand and a minute hand of a clock at right angles during their motion from 1.00 p.m. to 10.00 p.m.?**

(a) 9
(b) 10
(c) 18
(d) 20

**Ans. (c) :** In one hour, hour hand and minute hand are at right angles 2 times.
Time = 10 p.m – 1 p.m = 9 hr.
$\therefore$ No. of times, when both hands are perpendicular to each other in 9 hr = 9 × 2 = 18.